SYMPRAXIS CONSULTING

Extending Microsoft 365: Exploring the Art of the Possible

Julie Turner

Setting Expectations

- ✓ This is a 100 level INTRODUCTION to extending the Microsoft 365 SaaS platform
- ✓ We are NOT going to show/review code
- ✓ We are going to talk about all the ways to extend the platform, what tools you might need, and what scenarios you can accomplish
- ✓ Reference materials will be provided that tie back to each slide for future reference.
- ✓ *This is a firehose of information!*

SYMPRAXIS CONSULTING



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Julie Turner

Partner / CTO · Sympraxis Consulting

- Working with SharePoint since **2007**
- **Microsoft MVP**, Office Apps & Services since 2017
- M365 & Power Platform **Community Team** since 2019
- Open-source co-maintainer: **PnPjs** & **hTWOo**
- Co-host: ***Code. Deploy. Go Live.***

Agenda

- What is extensibility?
- Out of the Box Extensibility
- Low Code Extensibility
- Administration Tasks – Scripting
- Client-Side Extensions
- Community Resources

- *Server/Cloud Resources*
- *Authentication*
- *Design Frameworks*

What is extensibility?

Adjective
capable of being extended.

Extend - Verb
to increase in length, area, scope, etc.

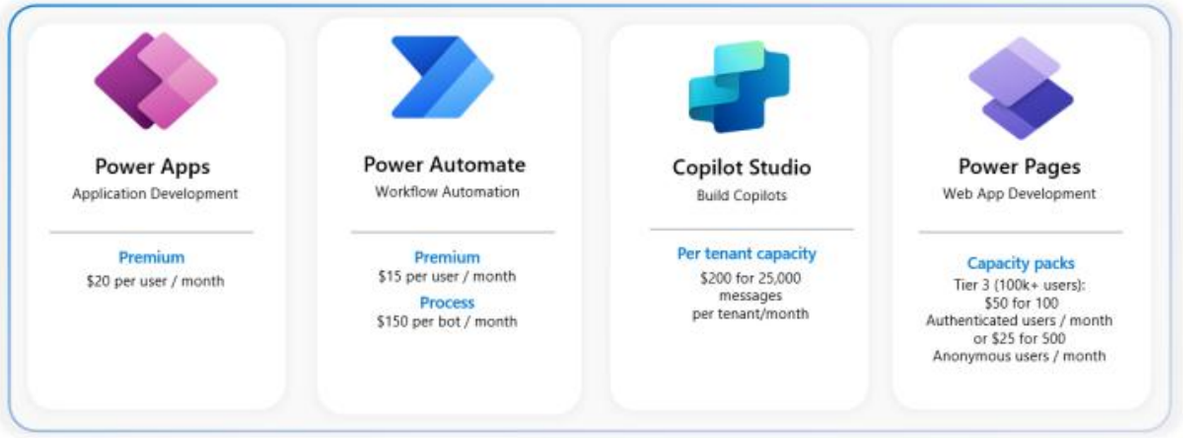


Out of the Box Extensibility

Templates

- Content Types and Document Templates
- Page Templates
- Site and List Templates
- Teams Templates
- List Formatting

Low Code Extensibility



PowerPlatform Connectors

Pair with CoPilot studio to allow others to create their own CoPilot experiences

LogicApps

- PowerAutomate base engine
- Standard/Consumption pricing
- Premium connectors billed differently

Pros

- Good for single entity business processes

Cons

- Cost to build/maintain + licensing (depending on connectors/platform)
- Complex logic can be hard to manage
- Prone to random failure

Administration Tasks - Scripting

One-time or infrequently performed tasks

- PowerShell – Various Microsoft 365 Modules
- Command Line Interface
  - Azure CLI
- Community Tools
  - PnP PowerShell
  - CLI for Microsoft 365

Client-Side Extensions

- Browser Based
  - Platform Hosted
    - SPFx
  - Self-Hosted
    - Teams Apps
    - Office Add-Ins (don't confuse with SharePoint Add-Ins)
- Native Mobile – iOS/Android
- Desktop Clients – iOS/Windows/Android

Tools – Client-Side Surfaces

- SharePoint
  - SPFx - SPPKG (Zip) files include code and manifest
  - Platform hosted (by default)
  - Supports SharePoint, Teams Tab/Personal App, Adaptive Card Extensions (ACE)
- Teams / AI Agents
  - Agents Toolkit
    - Teams AI V2 SDK – replacement for Teams FX library
      Bot Service interface for Teams
    - M365 Agents SDK – Bot Service interface for multiple services.
  - Self-hosted
  - Unified manifest for Microsoft 365 (Meta OS)
- Office Add-Ins
  - Office Add-ins XML manifest
  - Unified manifest for Microsoft 365 (Meta OS)
- Other
  - Microsoft365.com -> Unified Manifest (Meta OS)
  - Fluid Framework -> Azure Fluid Relay
  - LiveShare SDK – Leverages Fluid Framework and configured version of Azure Fluid Relay (Teams Meetings Only)
  - SharePoint Embedded – headless SharePoint storage with all the benefits of M365 functionality

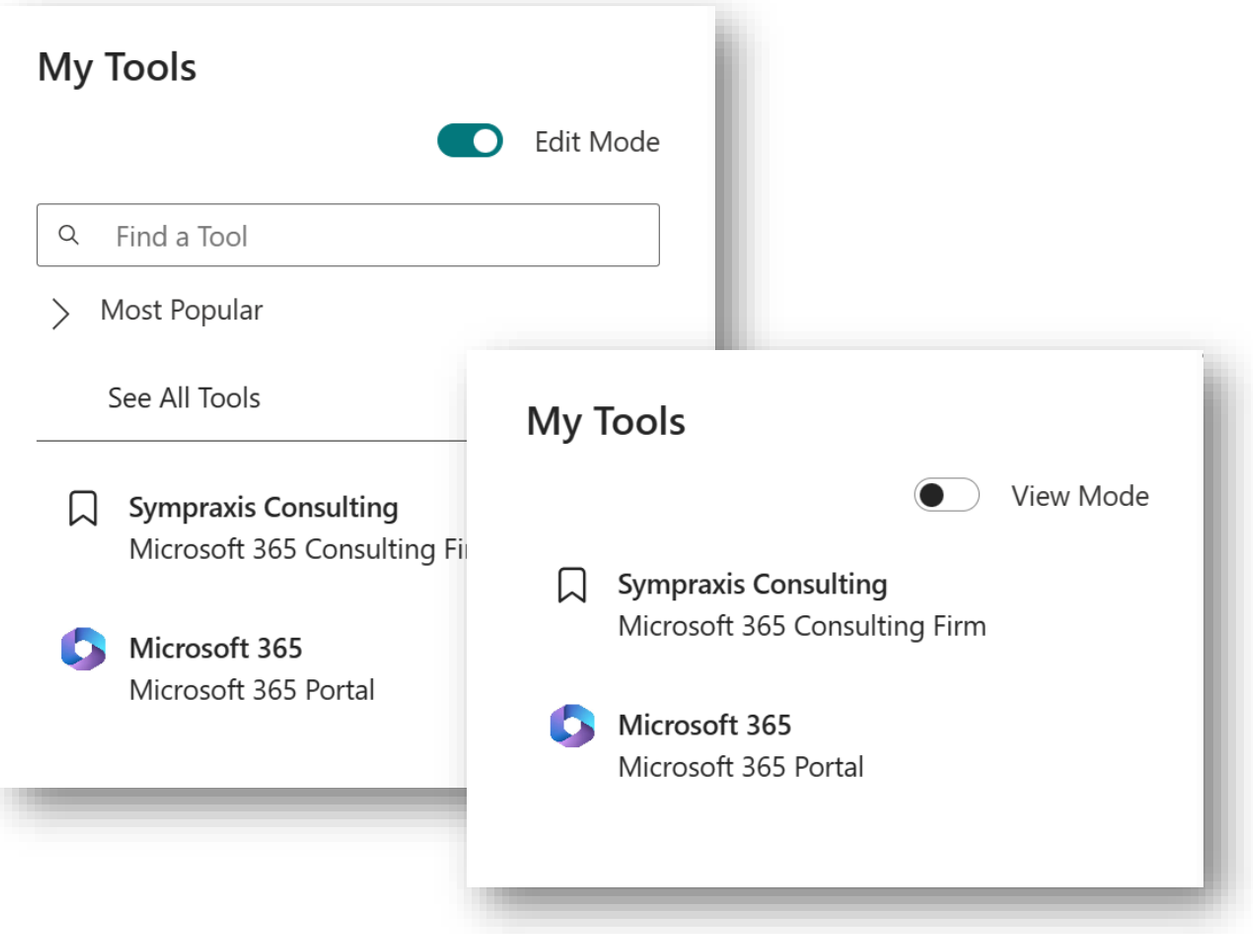
Delivery

- Microsoft App Source (App and Services listings)
  - Microsoft 365 Store – Integrated Apps (Teams, Microsoft365.com, Office Add-Ins Only)
  - SharePoint Store
- SharePoint App Catalog (SPFx/Teams, Outlook Add-Ins)
- Teams App Catalog (Teams)
- Teams Side Loading (for development)

SPFx – Web Part

Create customization snippet that can be added (1->many times)

- DIV Object
  - Surfaced anywhere in page that 1st party web parts can be added
- UX Scenarios
  - In page context user tools
  - Surface information from other sites
  - Surface information from external sources
  - Create mini-apps - aka Business Process Automation
- Can also surface as Teams Tab, Meeting App*, Chat* or Personal App

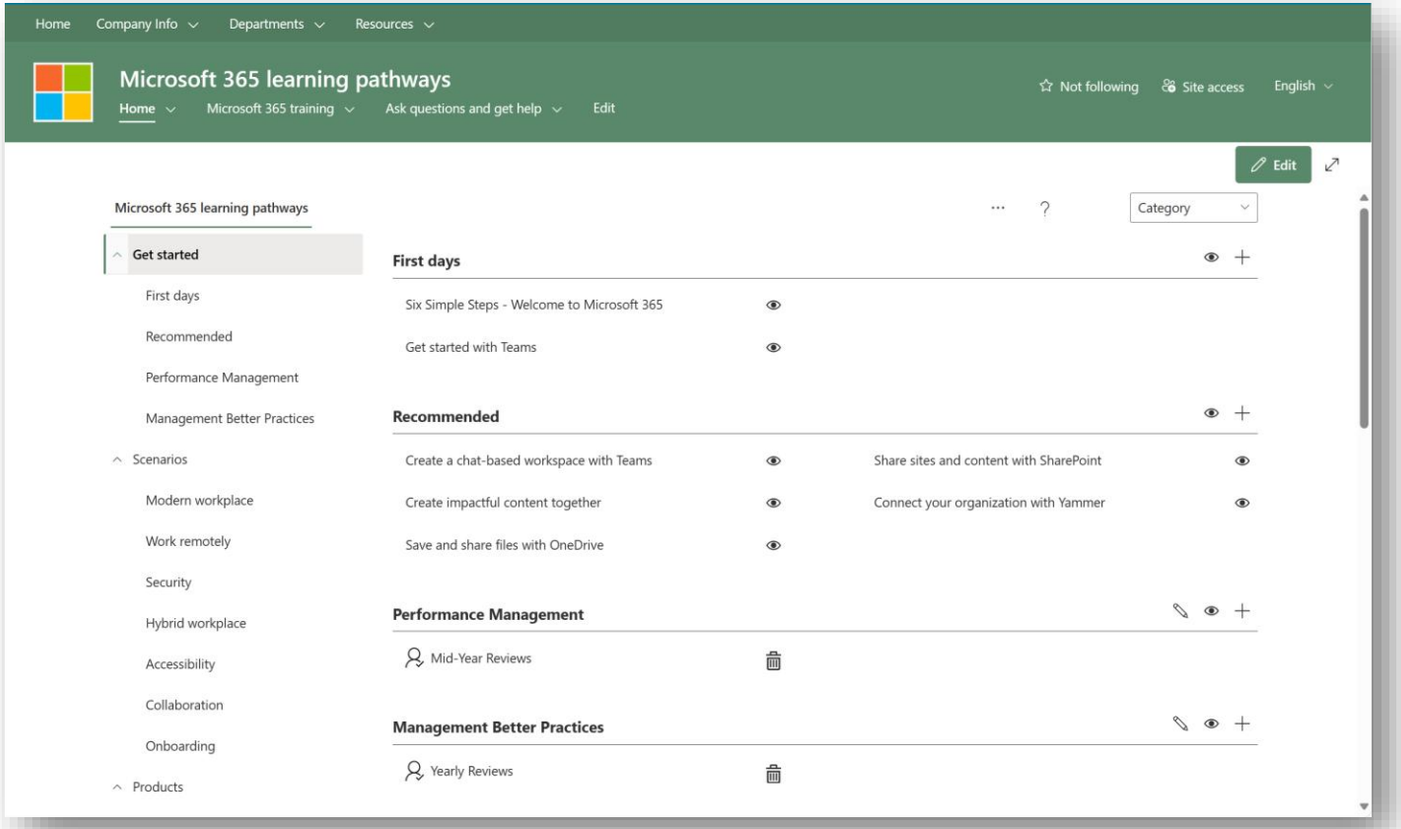


* Limited for guest users

SPFx – Single App Part Pages (special web part)

Ability to create page with embedded full-page webpart in site.

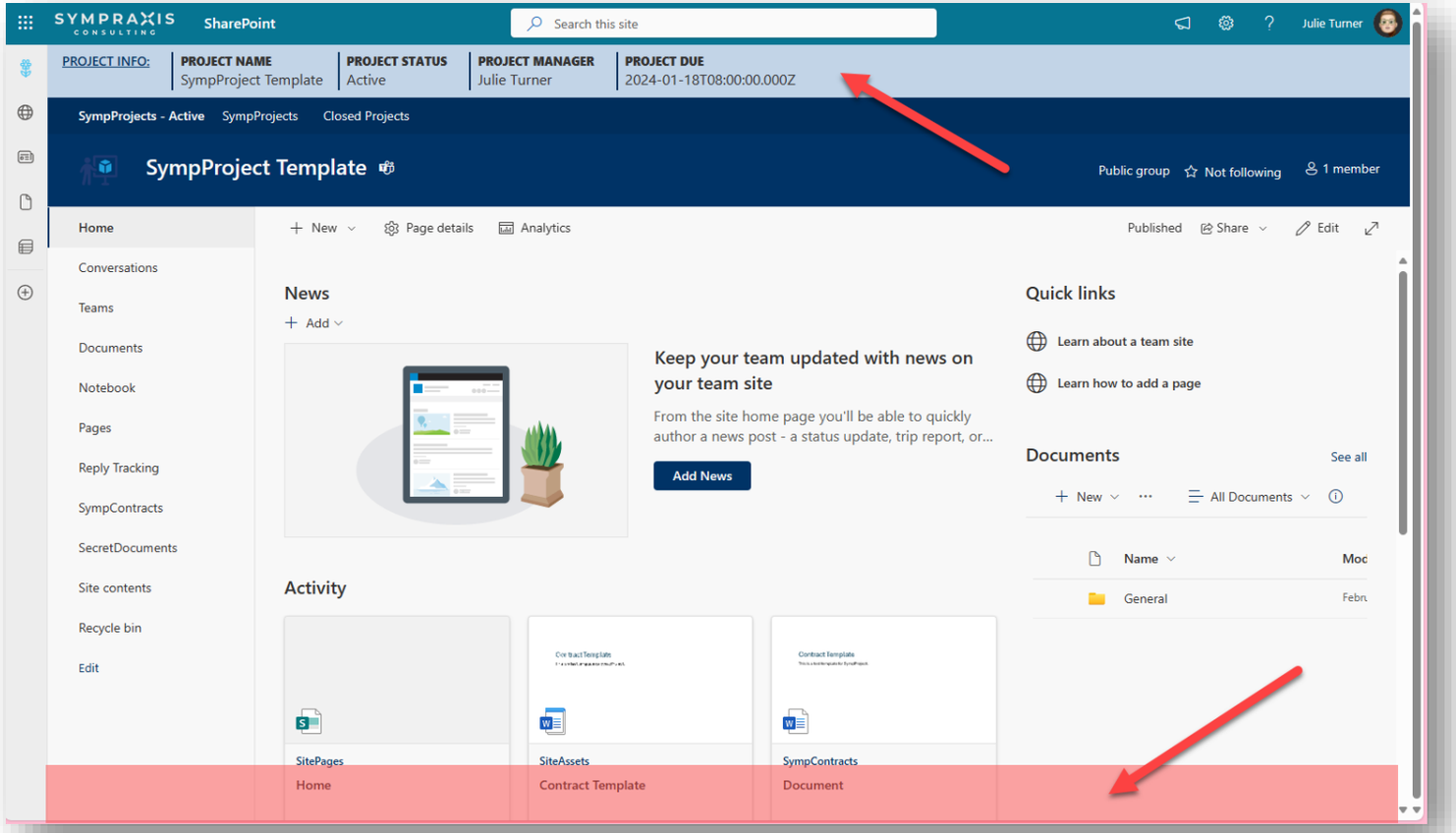
- DIV Object
  - Single column
  - No page toolbar
  - One web part
- UX Scenarios
  - Single Page Application
  - Full solution scenarios
  - Requires full page experience
- Extend UI: use an HTML modal dialog element



SPFx – Application Customizers

Runs on every page in the site/tenant unless you code it otherwise

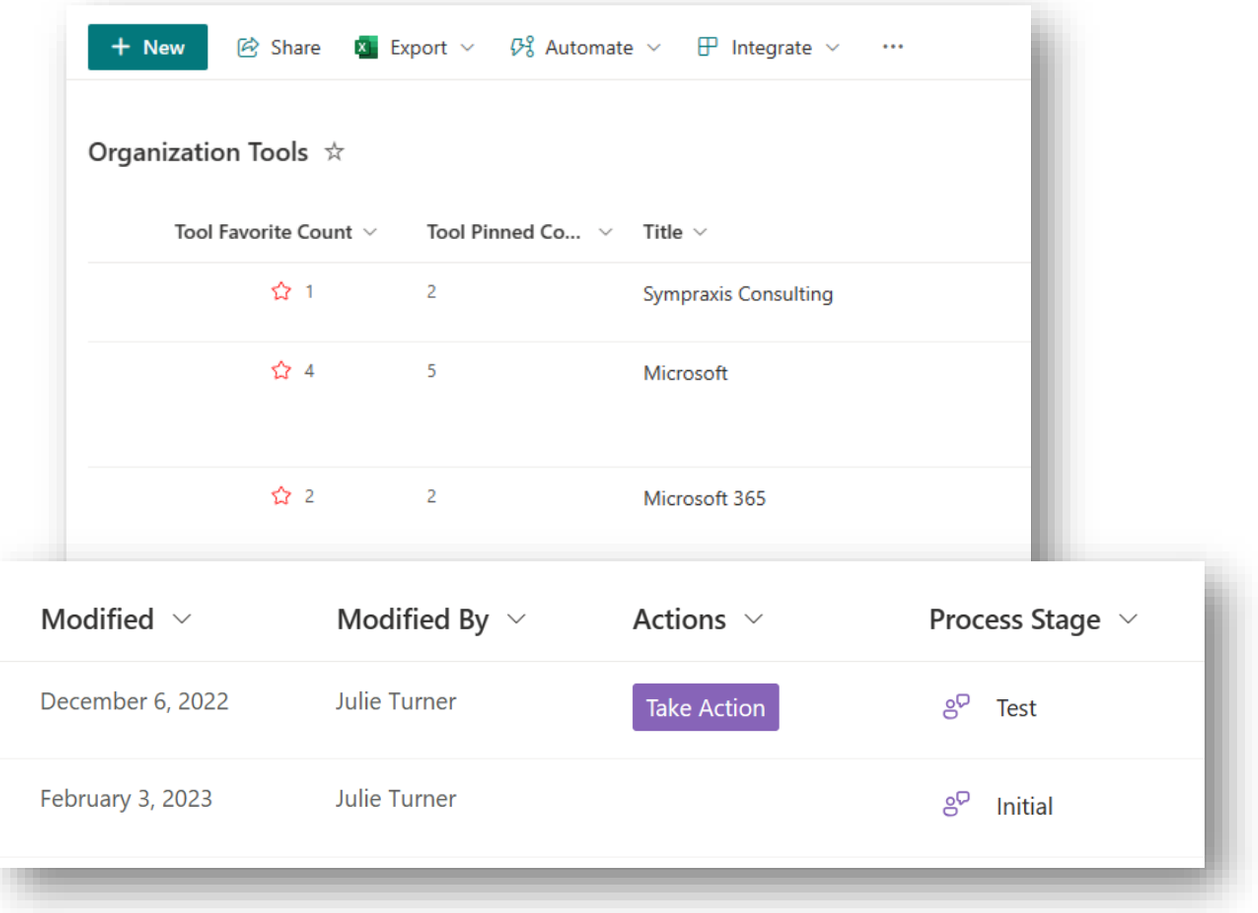
- DIV object – in page
  - Header (between suite bar and hub navigation)
  - Footer (pinned to viewport)
- UX Scenarios
  - Site/Tenant specific information
  - Interactive Site-specific functionality
- Scenarios – No UX
  - Analytics
- Extend UI: use an HTML modal dialog element



SPFx – Field Customizers

Runs on every page in the site/tenant unless you code it otherwise

- DIV Object
  - Replace the rendering of the field, with or without the fields value
- UX Scenarios
  - Transform the value
  - Provide extra visualizations
- Extend UI: use an HTML modal dialog element



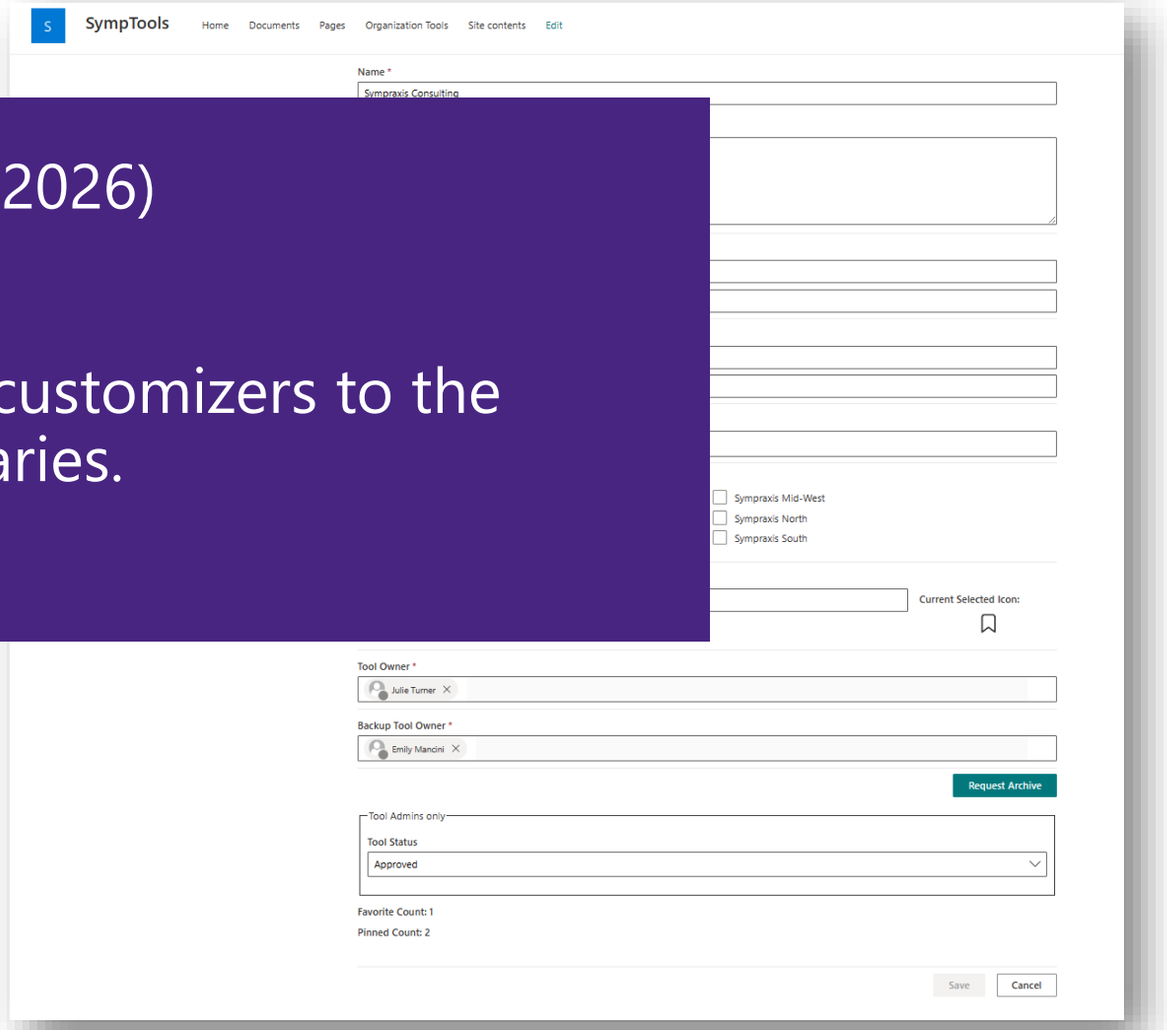
SPFx – Form Customizers

Customize the new, edit, view form for a SharePoint list

- Full Page Experience
- Build ANY lo
- Go extra mil
  custom secu
- UX Scenario
  - Complex data entry
  - Security scenarios
- Extend UI: use an HTML modal dialog element

SPFx 1.23.1 (mid June 2026)

Ability to target form customizers to the dialog in lists and libraries.



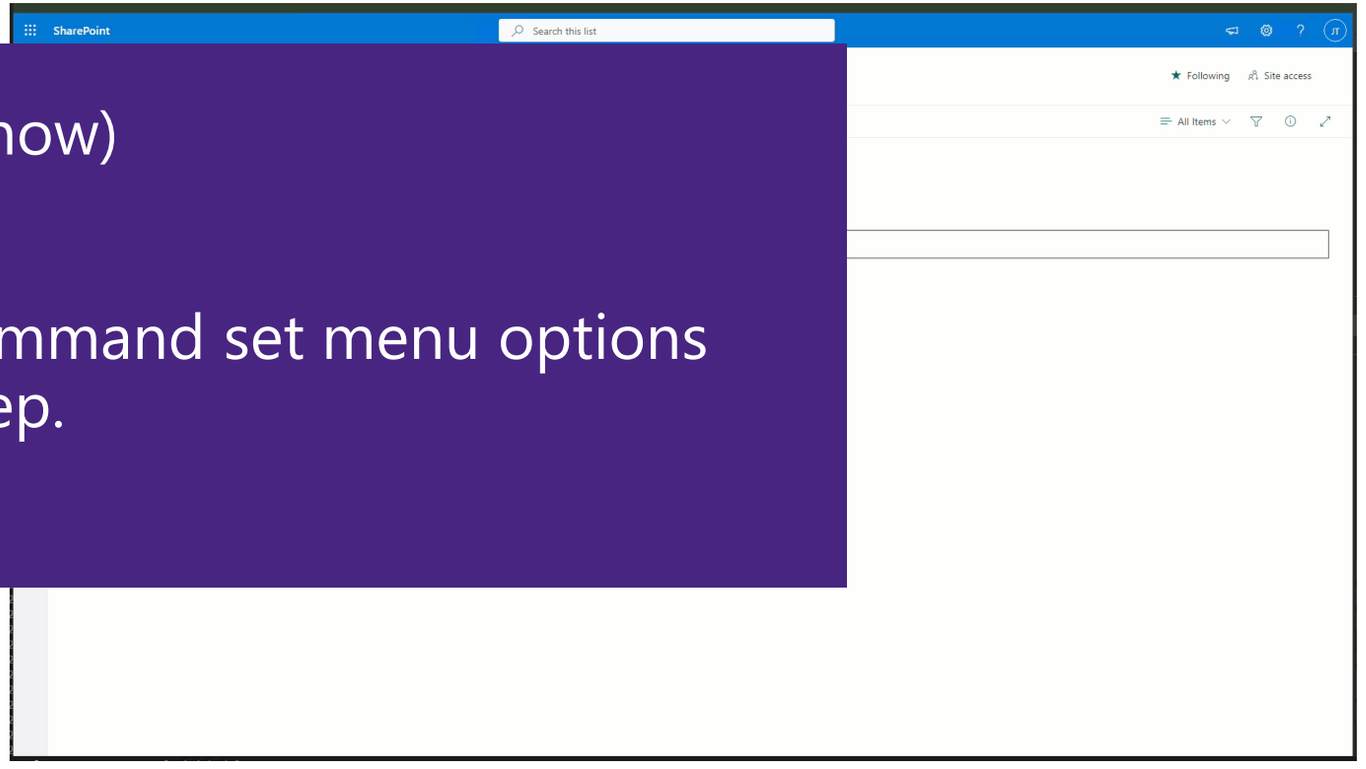
SPFx – Command Set Customizers

Runs on every list/library in a site/tenant unless you code it otherwise

- DIV Object
  - In toolbar or c
  - Provides "butt
    item context
- UX Scenario
  - Trigger action
  - Trigger action
  - Trigger actions for multiple items
- Extend UI: use an HTML modal dialog element

SPFx 1.23.0 (out now)

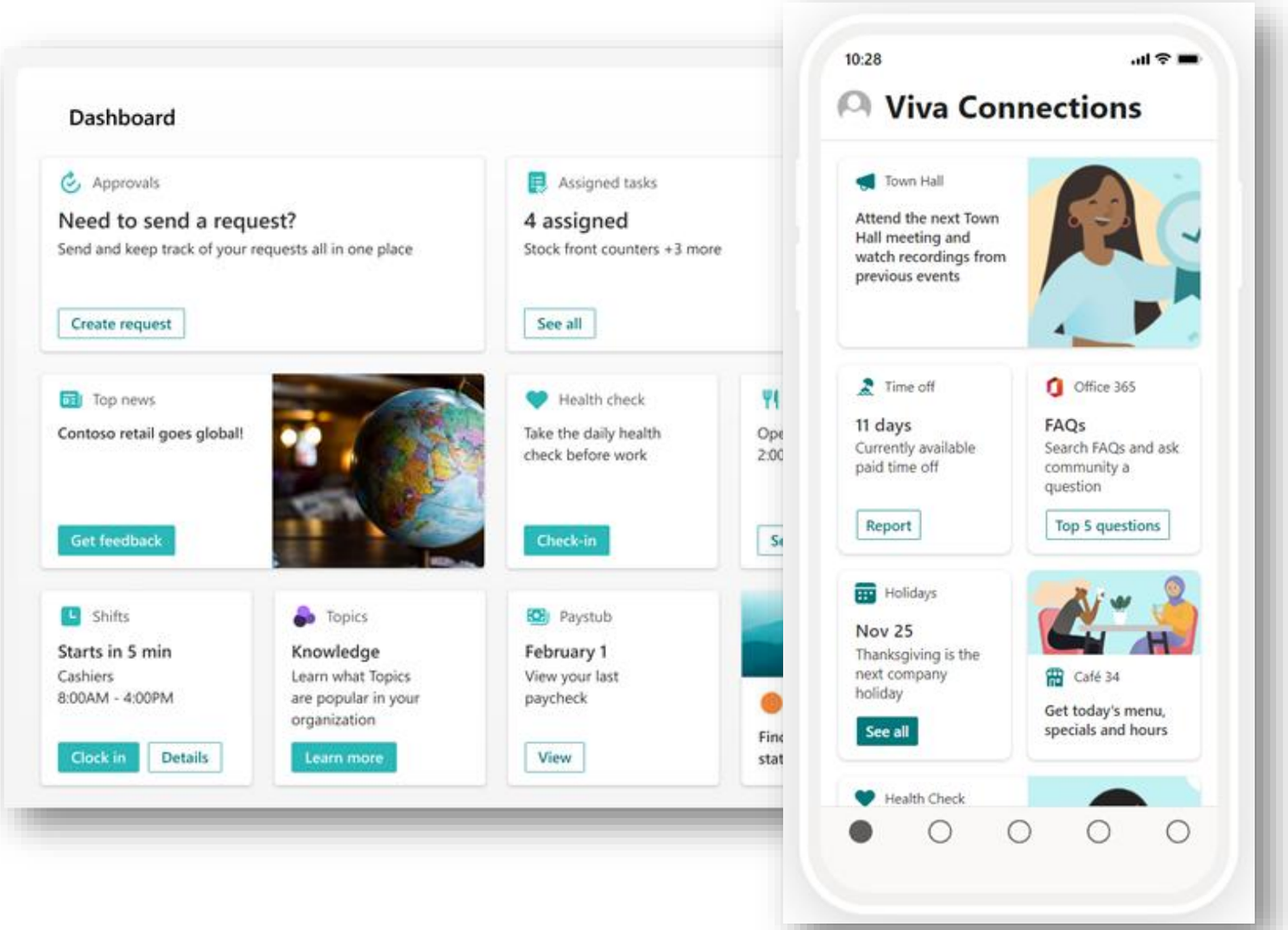
Ability to nest command set menu options up to 3 levels deep.



Viva Connections Adaptive Card Extension (ACE)

Optimized Adaptive Card based UX scenarios for job-related tasks

- Adaptive Card(s)
  - Med/Large Card View
  - Quick View “action card(s)”
- Optional full HTML/CSS
- Optimized for mobile
  - “Home Site” Dashboard
  - Surfaced in Teams mobile
- UX Scenarios
  - User based notifications
  - Business Process tasks
  - Announcements



ADVENTURE WORKS

Together We Innovate Adventure

Ready for Adventure

Seattle, WA
51 °F Cloudy 65°/48° 05/22/2025 MSN Weather

Vienna, Austria
62 °F Cloudy 67°/51° 05/22/2025 MSN Weather

Boston, MA
48 °F Heavy rain 48°/43° 05/22/2025 MSN Weather

São Paulo, Brazil
69 °F Mostly cloudy 77°/62° 05/22/2025 MSN Weather

Your quick actions

See all

Customer Loyalty

Customer Loyalty Sign-Up

Sign up customers for AdventureWorks loyalty program.

Customer Email

My Goals



My PTO 2025



Check Store Inventory

Search stores for inventory by model number

Model Number

No Match

NA No suggestions found

Your Wellbeing

Your well-being is important to us!



See More

Top News

Exciting Job Opportunities: Store Associate Manager and Sales Associate + 8...



See all

Camping

Camping

Camping product specifications and user manuals

Fishing

Fishing

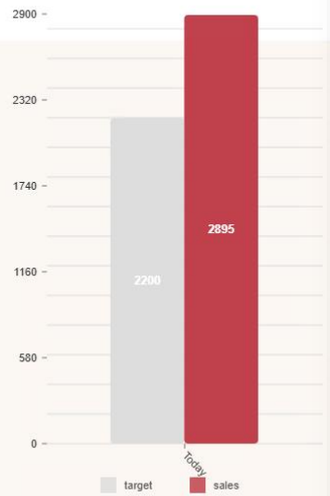
Fishing product specification and user manuals



You currently rank **3rd** in sales this month.

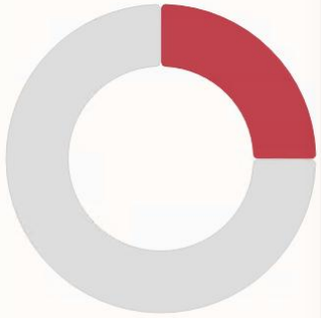
Sales Performance Overview

Daily | Weekly | Monthly

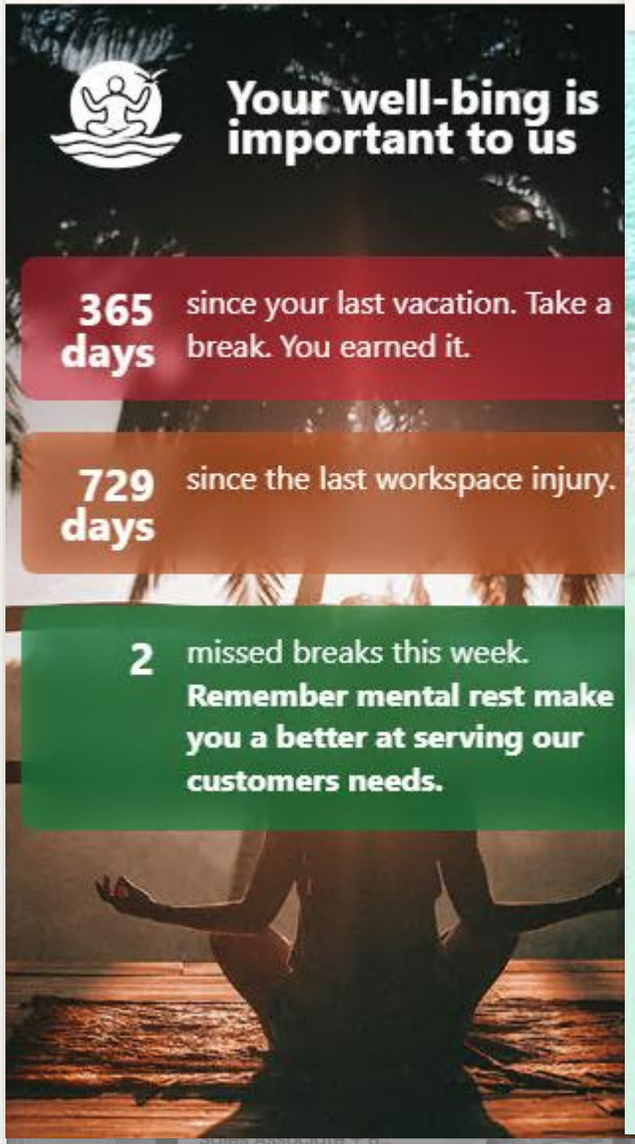


Loyalty Signup

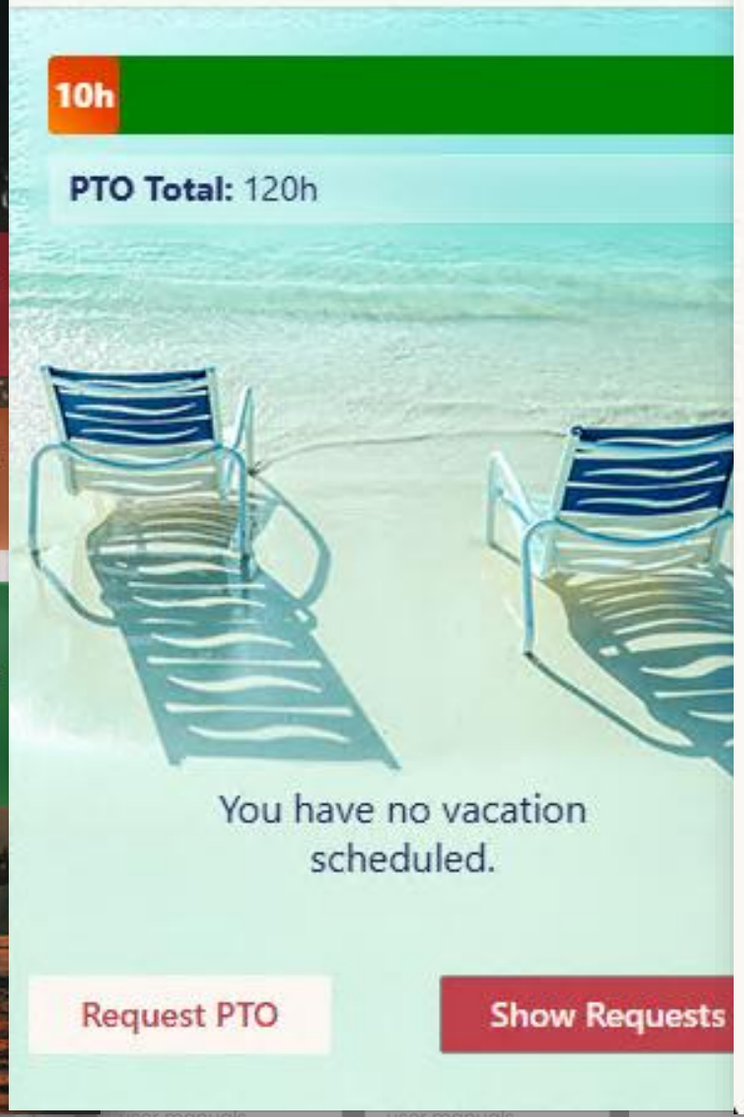
5	25%	20
Current	Complete	Target



Your Wellbeing



My PTO 2025





pp | Search

PP-SP-3030

Peak Performance Freestyle Ski Poles

Store	Qty
Chicago Adventure 675 N Michigan Ave Chicago, IL	28
Philadelphia Adventure 1625 Chestnut St Philadelphia, PA	22

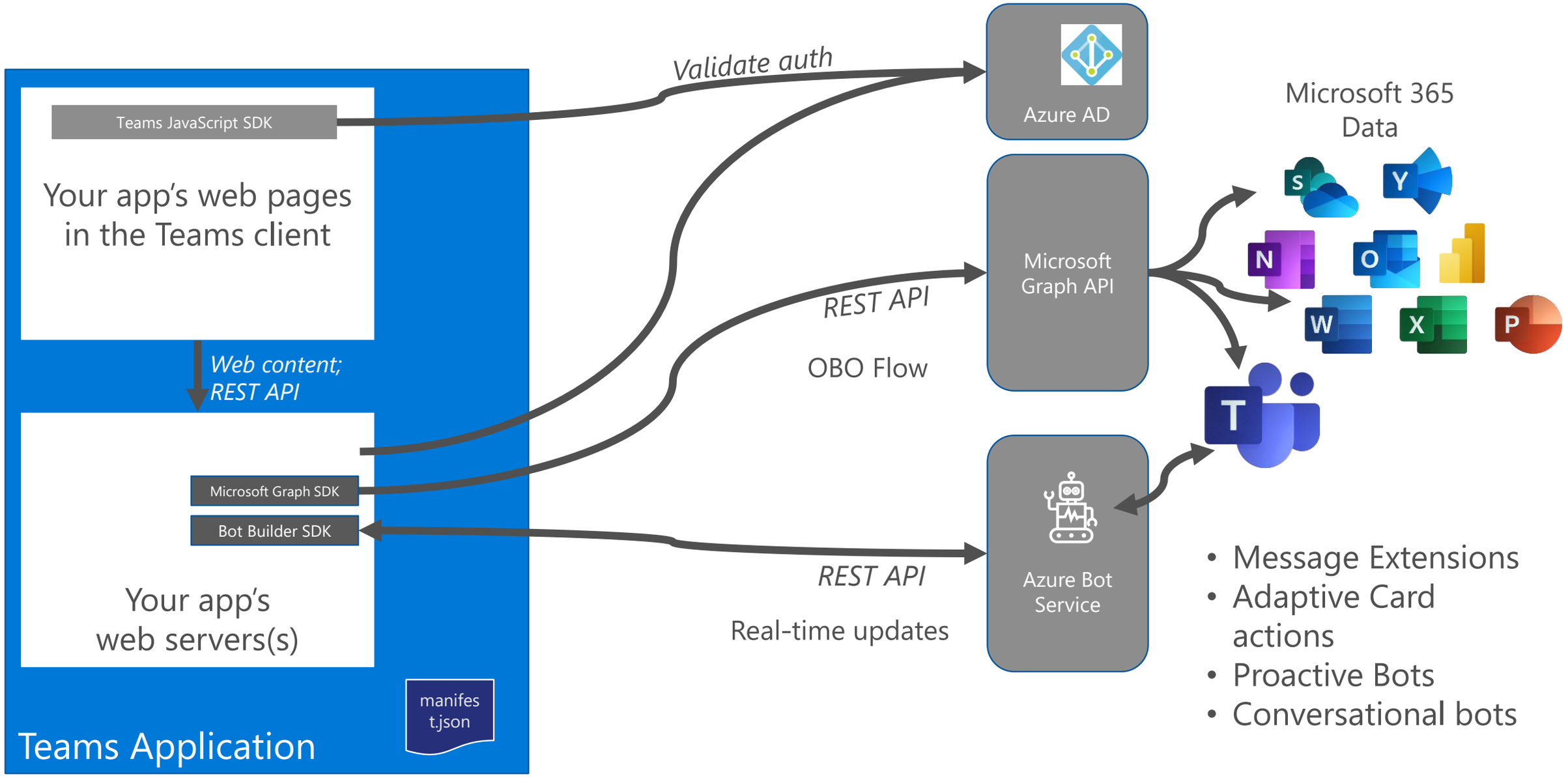
PP-S-4050

Peak Performance Cross Country Ski Specifications

Store	Qty
Chicago Adventure 675 N Michigan Ave Chicago, IL	22

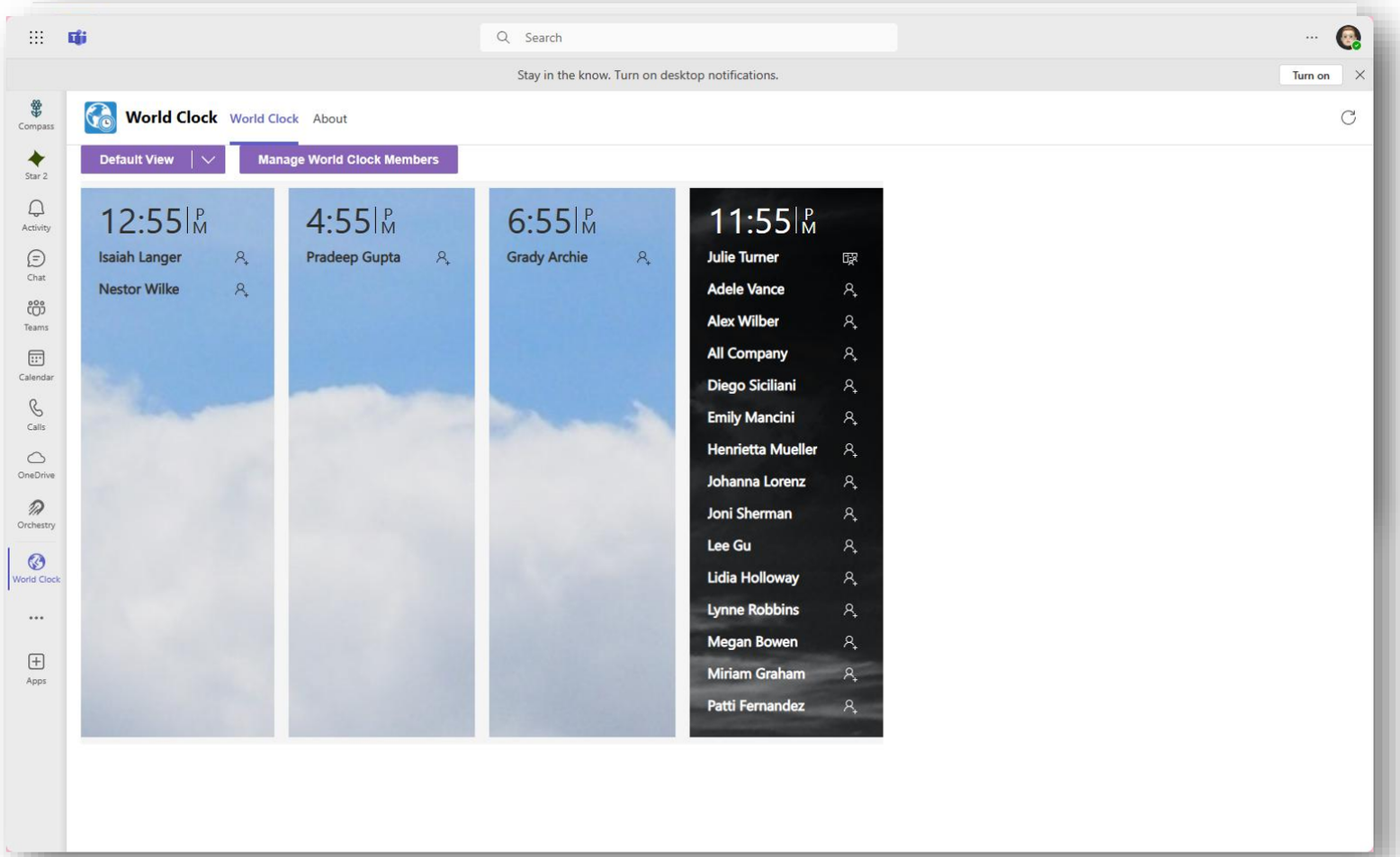
PP-1001

Teams Development



Teams - Tabs

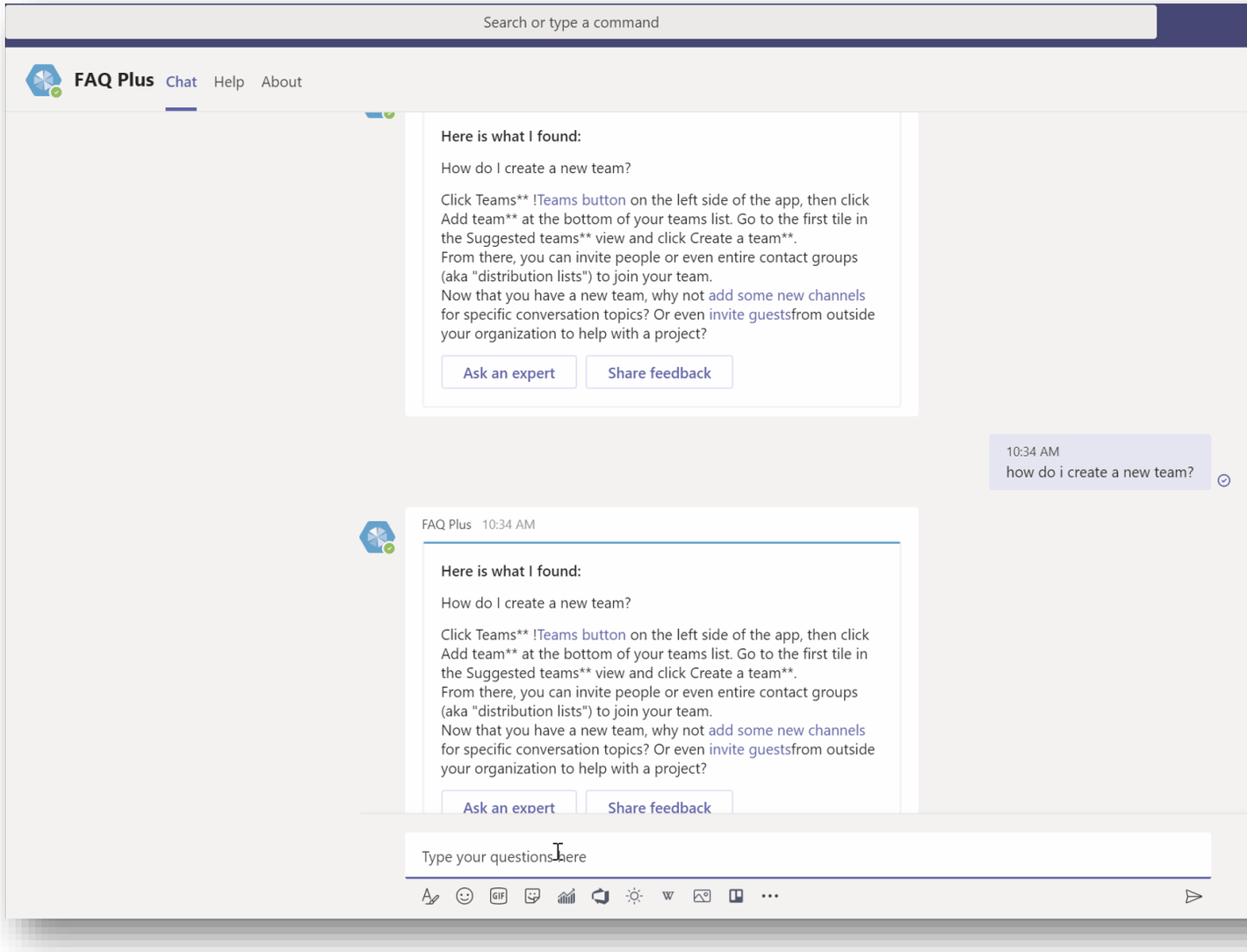
- iFrame your externally hosted web application in Teams
- Channel/Group Chat/Meetings Tabs are configurable
- Personal Tabs are not configurable
- Can also be deployed to SharePoint



Teams - Bots

Also known as chatbot or conversational bot.

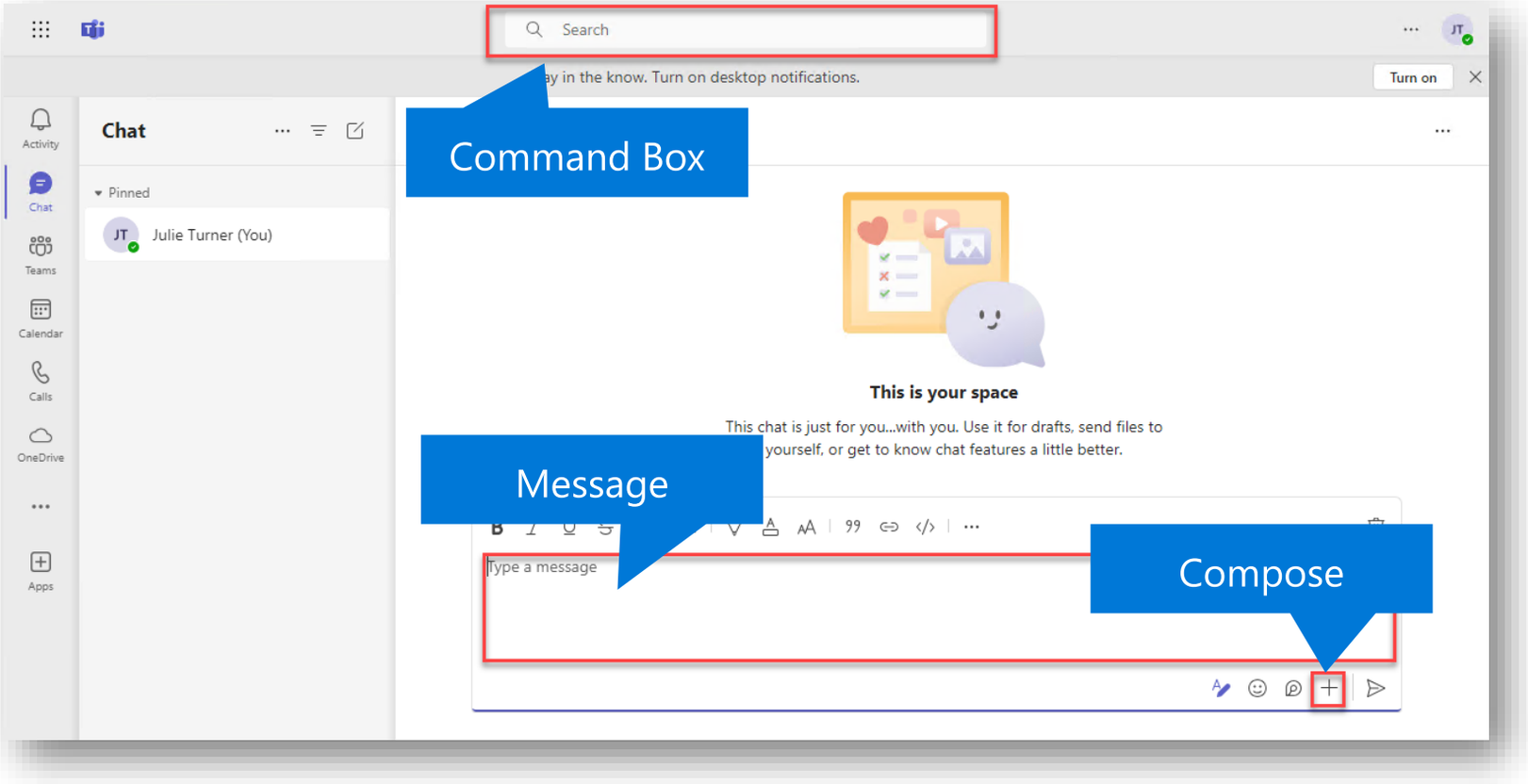
- Uses Bot Framework
- Conversational bots allow users to interact with your web service using:
  - Text
  - Interactive cards
  - Dialogs
    (referred as task modules in TeamsJS v1.x)



Teams – Messaging Extensions

Interact with web service through buttons and forms within the Microsoft Teams client

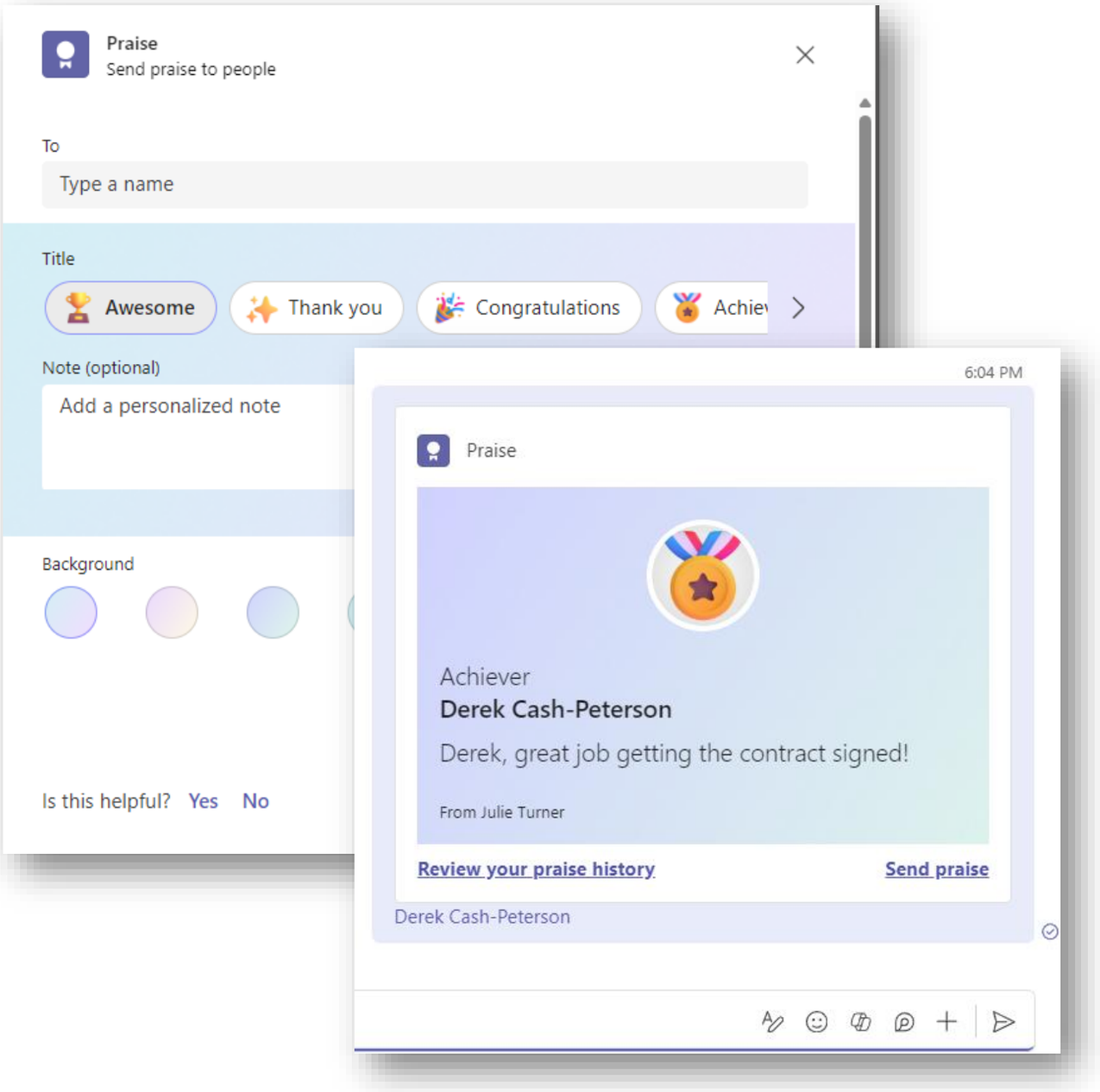
- Bot Framework
- API Message Ext
  - call your REST API directly.
  - limited functionality
- Actions
- Search
- Link Unfurling



Teams – Actions

Interact with user in chat/post UX

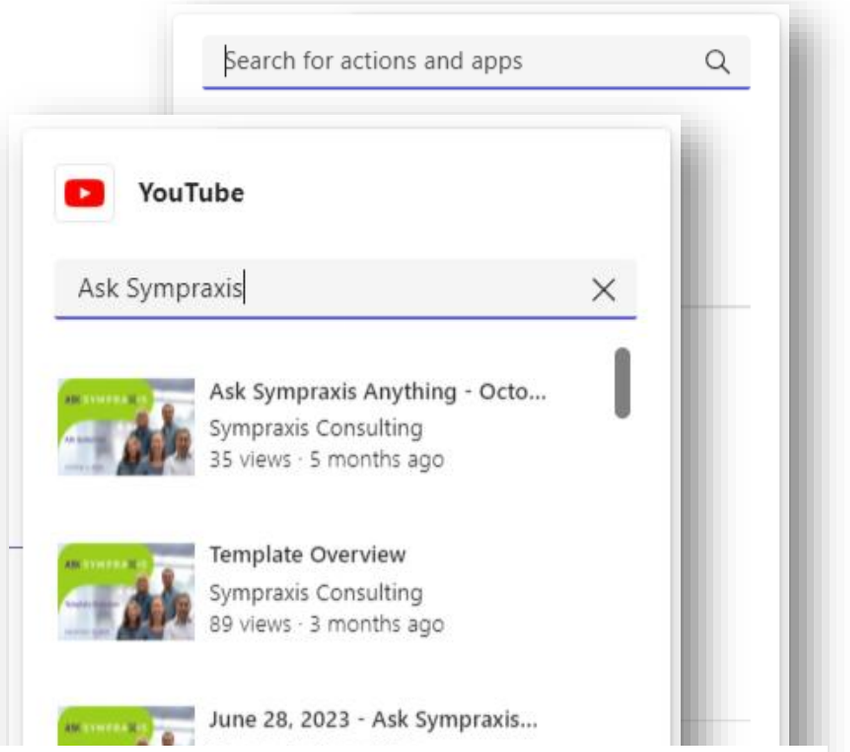
- Present the users with a modal pop-up to collect or display information
- Service responds by inserting a message into the conversation directly
- Multiple forms can be stitched together to behave like a “wizard”
- Invoke actions from compose message area, the command box, or inside the message (existing message sent to the service)

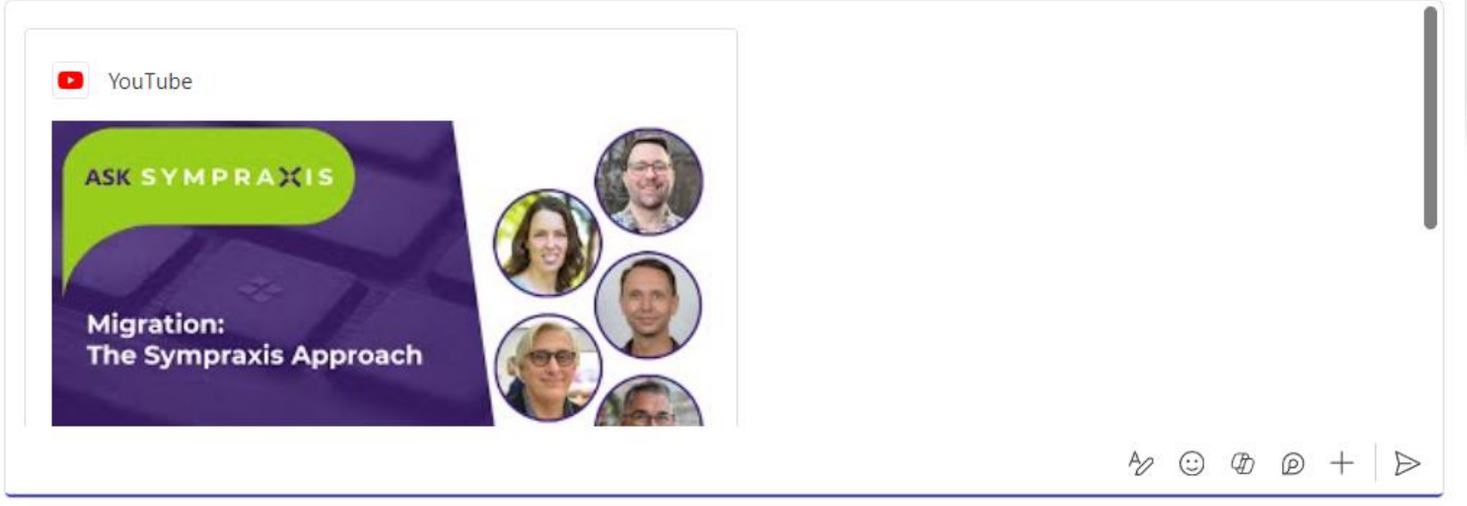


Teams – Search

Provide user with custom search results

- Invoke message includes the search string submitted by the user
- Respond with a list of cards and card previews
- Trigger from command b or compose message bu not the message itself.

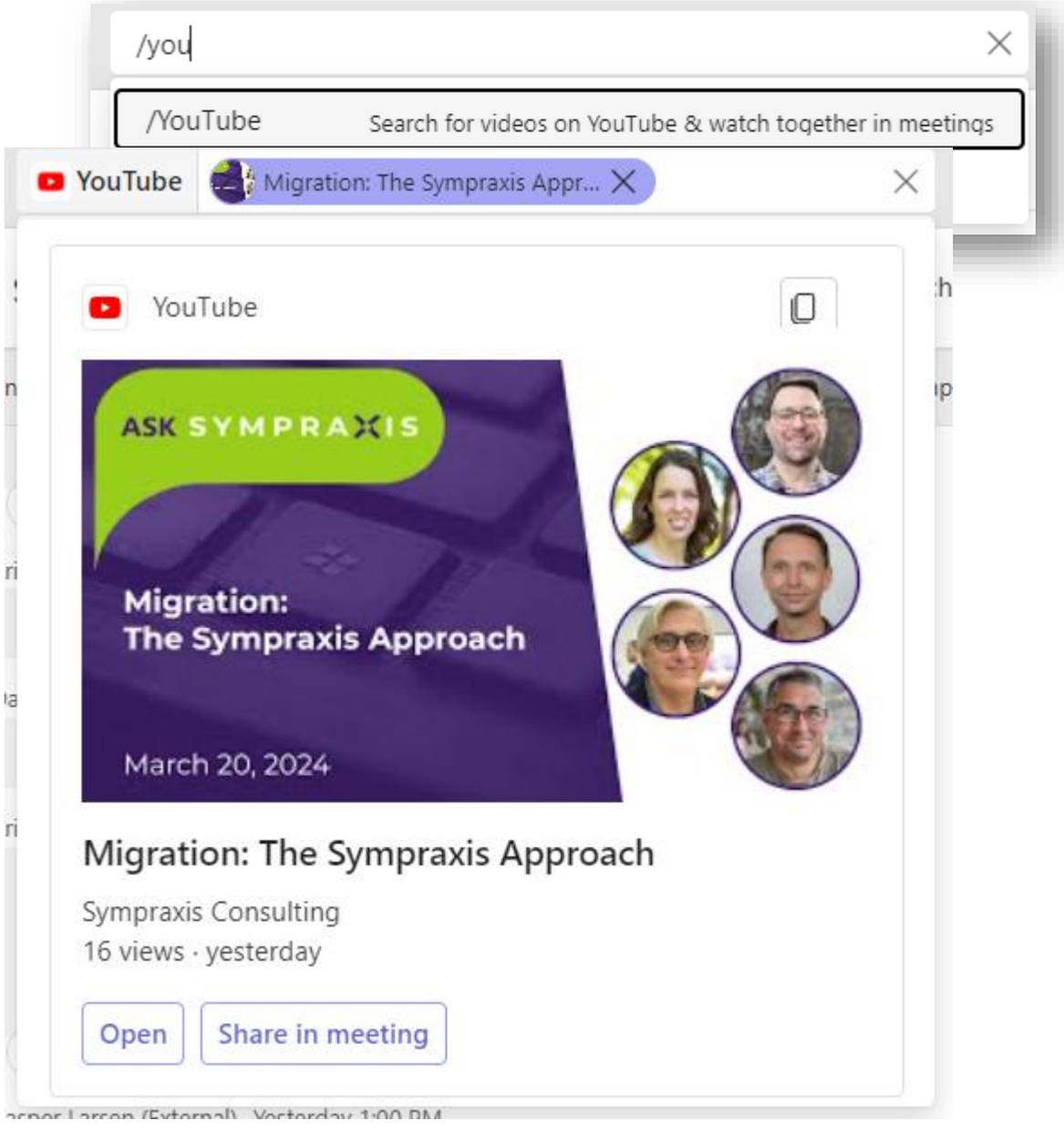




Teams – Search

Provide user with custom search results

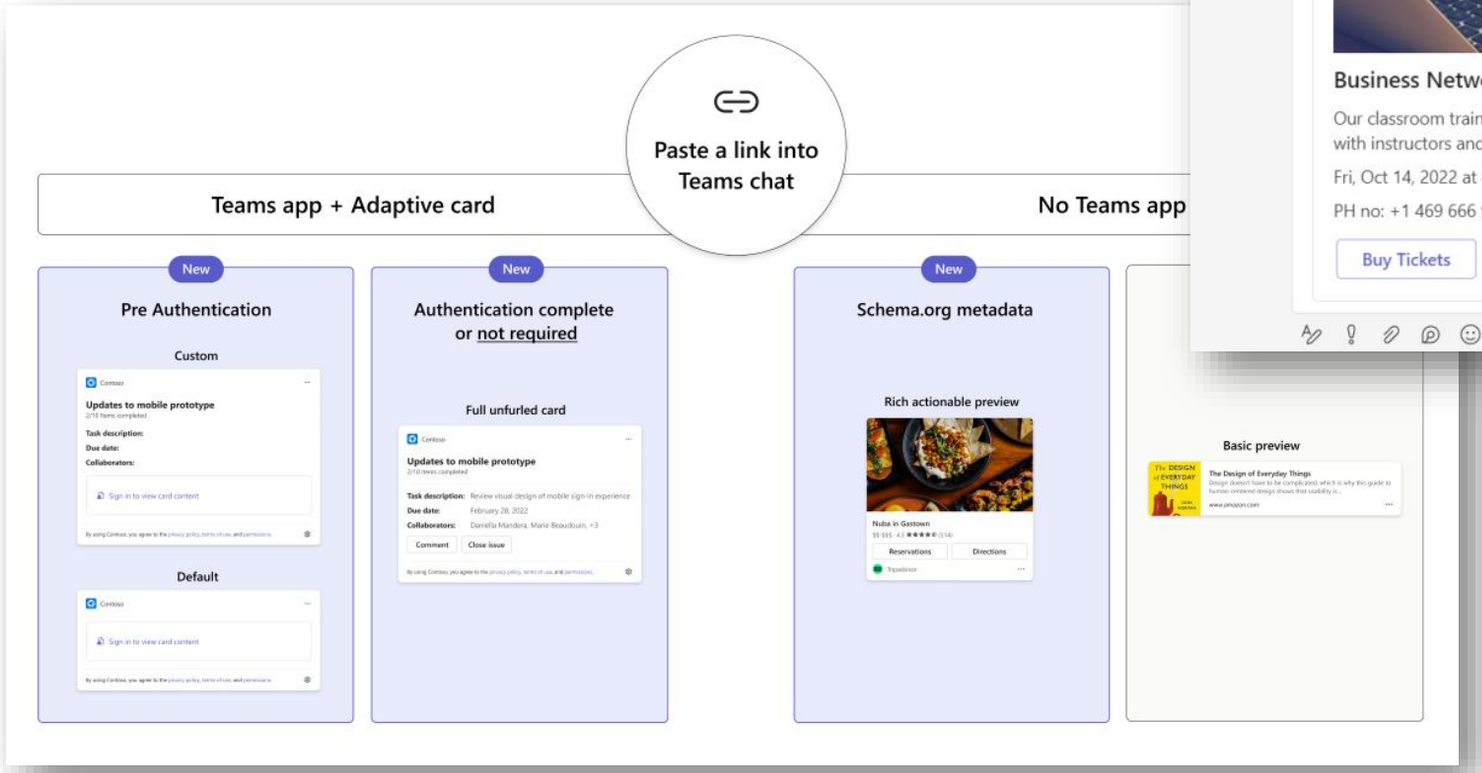
- invoke message includes the search string submitted by the user
- respond with a list of cards and card previews
- Trigger from command box or compose message but not the message itself.

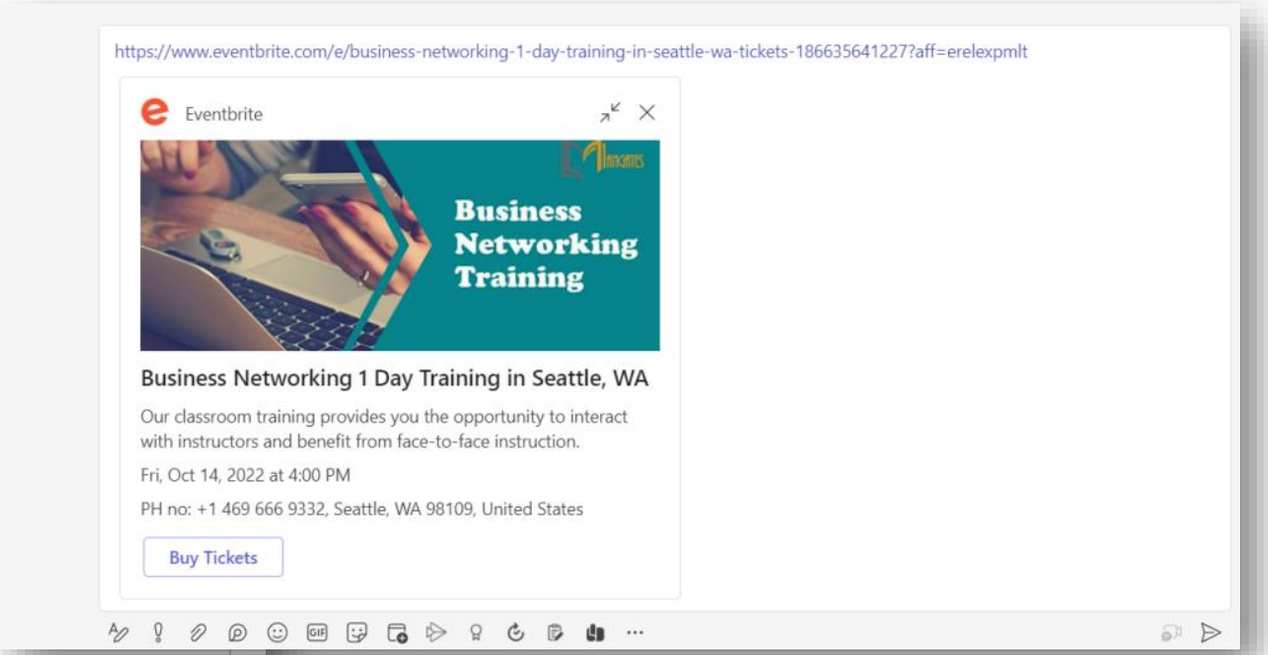


Teams – Link Unfurling

Provide user with customized adaptive card in response to links with specific host.

Can be enabled without having to have the user install the customization

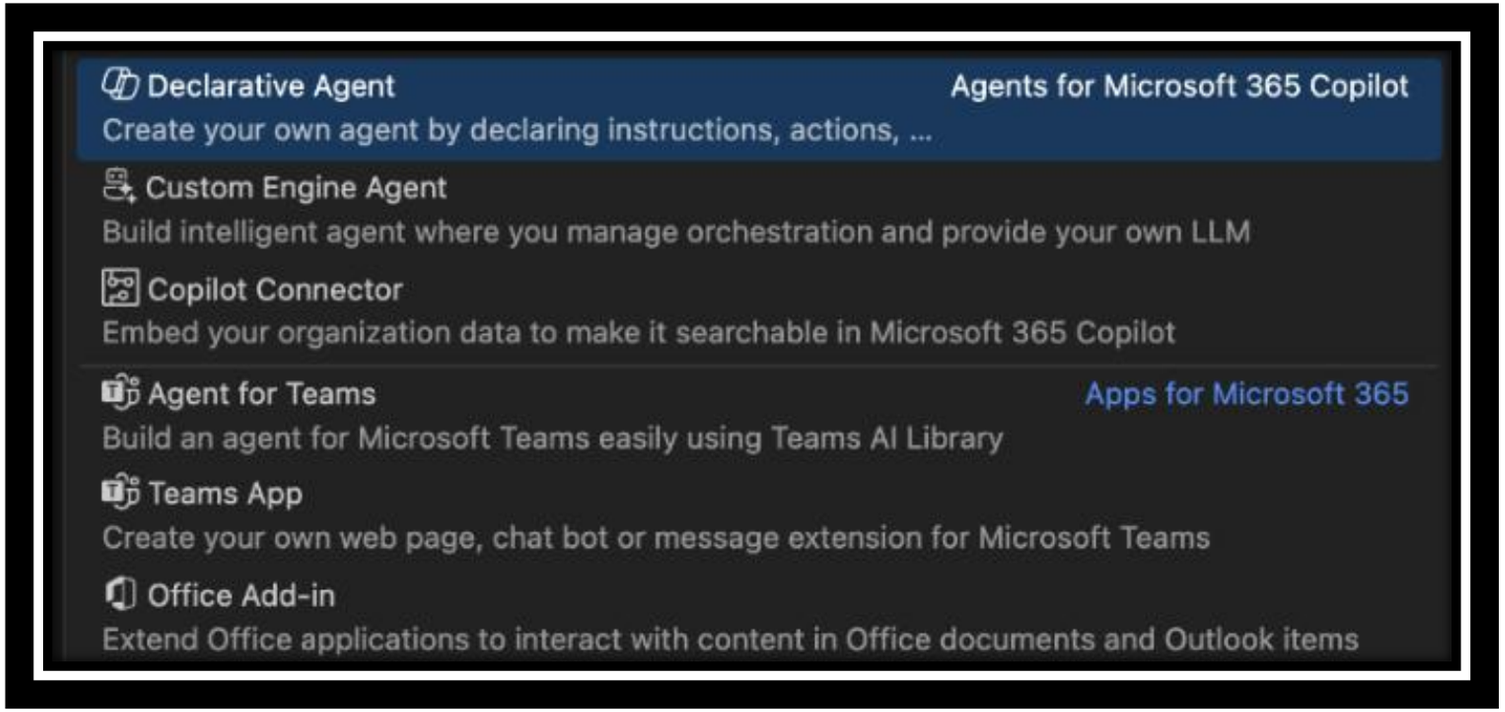




Works in desktop and mobile

Agent Development

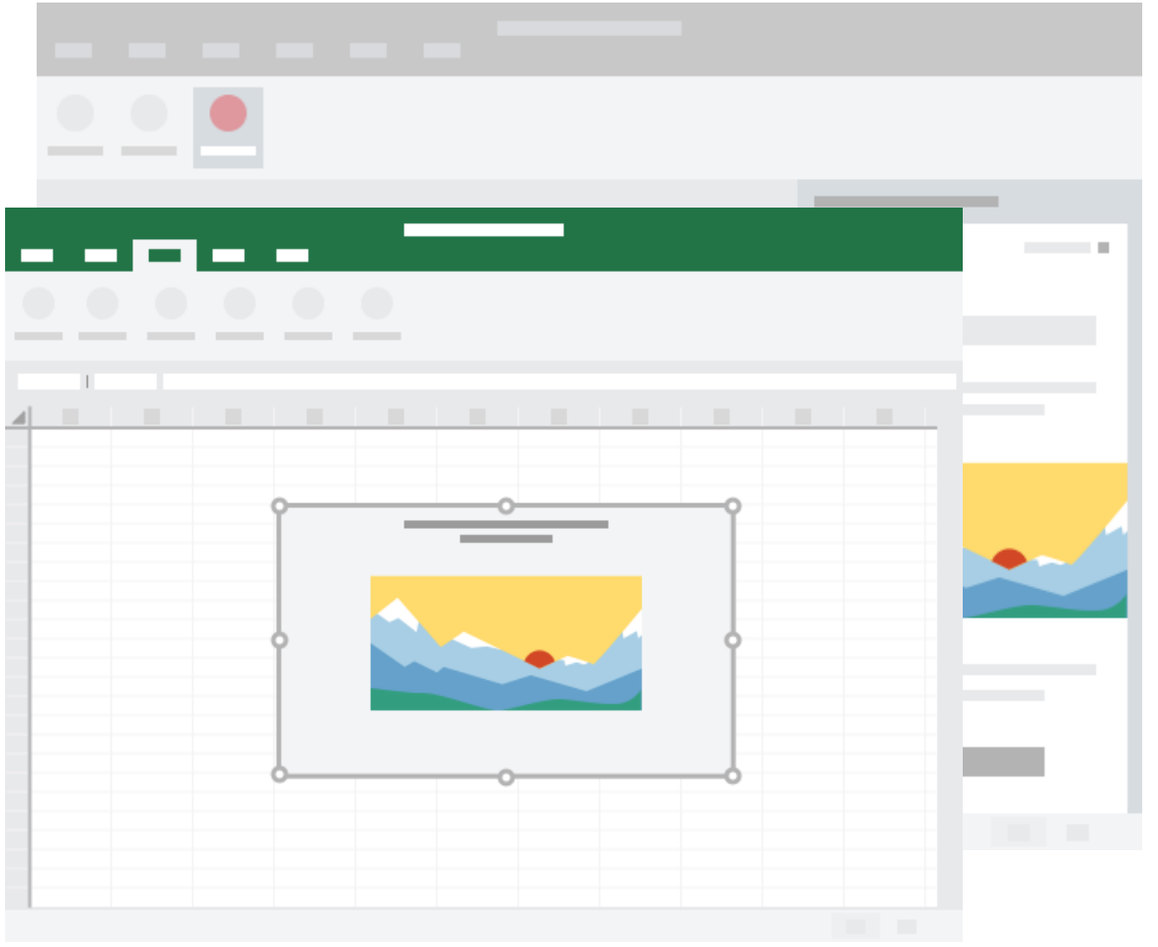
- Declarative Agents (json manifests)
- Custom Engine Agents (bring your own)
- Copilot Connectors (fka Graph Connector)
- Agent for Teams-only (fka Bot w/ Teams AI V2 SDK)



Office Add-Ins

Extend Office Client functionality (Outlook, Word, Excel, PowerPoint, OneNote, Project, Visio)

- Custom ribbon buttons and menu commands
  - **collectively called "add-in commands"**
- Insertable task panes
- Embed Web-Based objects called “content add-ins” which can surface external content
  - **web pages, videos, etc**
  - **Excel/PowerPoint only**



Community Support

- Community Calls
- Videos
- Guidance
- Samples/Solutions
- SDKs
- Tools
- Extensions
- Forums



Resources

https://symp.info/m365-ext-links